Sample Exam Questions for SE

Score	Evaluator	Section I: **Completion by Matching.** Identify the letter of the choice that best complete each statement.

A. Availability
B. Activity
C. Client-server
D. Context
E. Development
F. Ethnography
G. Equivalence partition
H. Functional requirements
I. Heterogeneity
J. Interaction
K. Open-source
L. Pair programming
M. Pipe and filter
N. Reuse
O. Scenario testing
P. Sociotechnical systems
Q. Specification
R. System requirements
S. Technical systems
T. Test-driven
U. Use-case
V. User interfaces
W. User requirements
X. Validation
Y. Verification
Z. Workflow

_O__ 5. _____________ is an approach to release testing where you write a story describing how a system may be used and design tests based on the sequence of events in the scenario.

_P__ 6. ______________ is/are self-aware and include defined operational processes and procedures.

_A__ 7. ______________ is the ability of a system to deliver services when requested.

_L__ 19. Requirements expressed as scenarios, ___________, and test-first development are three important characteristics of extreme programming.

_W__ 20. _________________ are statements in a language that is understandable to a user of what services the system should provide and the constraints under which it operates.

Score	Evaluator	Section II: **Multiple Choices.** Identify the choice that best completes the statement or answers the question.

_D__ 21. ____ is the phenomenon where a module in a system implements or partially implements a number of different system requirements.

a. Concerning c. Floating
b. Scattering d. Tangling

_D__ 22. Which is **NOT** one of the 4 levels at which software reuse is possible?

a. The abstraction level d. The specification level
b. The object level e. The component level

c. The system level

_E__ 23. Which is **NOT** one of the principal system re-engineering activities?

a. Source code translation
b. Program structure improvement
c. Program modularisation
d. Reverse engineering
e. Program Debuging
f. Data re-engineering

_A__ 24. Which is **NOT** one of the four principal dependability properties?

a. Understandability
b. Availability
c. Security
d. Reliability
e. Safety

_B__ 25. Which is **NOT** one of the 4 sectors in each loop in Boehm's spiral model?

a. Objective setting
b. Feasibility study
c. Planning
d. Risk assessment and reduction
e. Development and validation

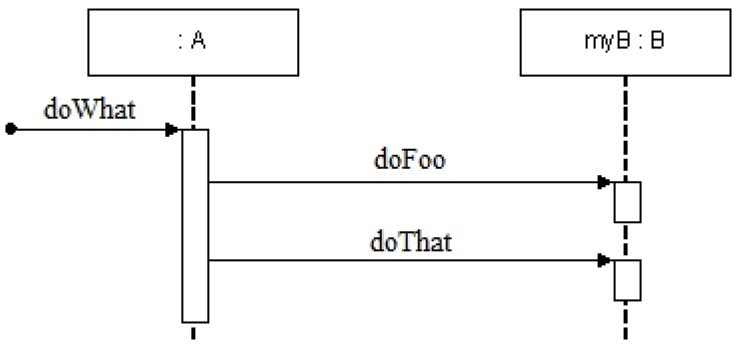
Score	Evaluator	Section III: **Short Answer**. Please **Select** (**mark a √ before the No. of the question chosen**) and briefly answer **2** from the following 3 questions. (Question 31~33, 9 *pts for each short answer, totally* 18 *pts*)

____ 31. Explain, using examples as appropriate, what is meant by the following architectural style (choose only one from the following four styles):

a. Pipes and filters architecture
b. Layered architecture
c. Client-server architecture
d. Object-oriented architecture

Score	Evaluator	**Section IV: Design Problems.**

32. Consider the following Sequence Diagram in design stage, draw a equivalent communication diagram and write down the corresponding code segments in any object-oriented programming language for class A and class B.



Score	Evaluator	Section V: **Essay.** Answer the following question.

35. (? related to the course projects)